

FINITE GROUPS THAT ARE THE PRODUCT OF EVERY PAIR OF NON-CONJUGATE MAXIMAL SUBGROUPS ARE SOLUBLE

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ABSTRACT. We prove that every finite group which coincides with the product of any two of its non-conjugate maximal subgroups is soluble, answering Problem 10.34 of the Kourovka Notebook (V. S. Monakhov, 1986) in the negative. The almost simple case was settled by Tikhonenko and Tyutyanov (2010); the obstacle to the general case was the socle S^k with $k \geq 2$, which no counting bound can control. We remove it with a divisibility criterion: a single pair of automorphism-stable conjugacy classes of subgroups of the simple group S , subject to a valuation inequality, excludes the socle S^k for *all* $k \geq 2$ and every admissible embedding at once. Such pairs are constructed uniformly for every infinite family of finite simple groups, with Zsigmondy primes as the arithmetic obstruction; the sporadic groups and all remaining small cases are settled by independently re-checkable certificates produced in GAP.

1. INTRODUCTION

Problem 10.34 of the Kourovka Notebook, posed by V. S. Monakhov in 1986 [15], asks:

Does there exist a finite non-soluble group that coincides with the product of any two of its non-conjugate maximal subgroups?

Soluble groups with this property exist in abundance: any group whose maximal subgroups are all conjugate satisfies it vacuously, and $S_4 = AB$ for every pair of non-conjugate maximals A, B . The content of the problem is whether the property forces solubility. We prove that it does.

Theorem 1.1 (Main Theorem). *Let G be a finite group such that $G = AB$ for every pair of maximal subgroups $A, B \leq G$ that are not conjugate in G . Then G is soluble.*

Throughout, we say G has *property P* if $G = AB$ for all non-conjugate maximal $A, B \leq G$.

The almost simple case of Theorem 1.1 was proved by Tikhonenko and Tyutyanov [28]. We use their theorem as the essential $k = 1$ branch of the main proof. The genuinely new burden is $k \geq 2$. Tikhonenko and Tyutyanov also record that Zenkov had announced a negative answer to the full problem in a 1997 conference abstract [32], while explicitly stating that no proof of that announcement was known to them. Searches completed on 1 August 2026 found no published full proof; accordingly, we claim a proof of the stated theorem, not absolute priority for the conclusion. The search record and its access limitations are in the supporting-materials source ledger. For comparison only, Theorem 4.3 records how the present criterion also recovers the almost simple conclusion; neither that theorem nor its family-by-family application is a dependency of the Main Theorem.

The maximal factorizations of finite simple groups and their automorphism groups were classified by Liebeck, Praeger and Saxl [18]. Their tables classify the factorizations that *do* occur.

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Our route needs only to exhibit, for each coordinate closure, one pair that cannot factorize; the valuation obstruction avoids importing the full maximal-factorization tables.

Strategy. A minimal counterexample G has a unique minimal normal subgroup $N = S^k$ with S non-abelian simple, and embeds in $\text{Aut}(S) \wr S_k$ (§2). For an automorphism-stable conjugacy class $[V]$ of suitable subgroups $V < S$, the normalizer $B_V = N_G(V^k)$ is a maximal subgroup of G of order exactly $|G/N| \cdot |V|^k$ (§3). Property P forces $G = B_V B_W$ for distinct such classes, and the resulting integrality condition $|B_V \cap B_W| \in \mathbb{Z}$ produces, for a well-chosen prime p , the divisibility $p^{dk} \mid |G/N|$ with d independent of k , which is incompatible with $|G/N|$ dividing $x^k k!$, where $x = |X/\text{Inn}(S)|$ for the relevant $X \leq \text{Aut}(S)$ (§4). The upshot (Theorem 4.1) is that a *single* pair of stable classes in S , satisfying a valuation inequality, rules out the socle S^k for every $k \geq 2$ and every admissible embedding at once.

The $k \geq 2$ branch thus reduces to exhibiting such a pair for every finite simple group S . By the classification of finite simple groups [10, Ch. 1, Table I, pp. 8–10] (see also [23, Thm. 1.1]), this is a finite list of infinite families plus 26 sporadic groups and the Tits group. For the infinite families the pairs are constructed uniformly in §6, with Zsigmondy primes of the relevant torus exponents as the obstruction primes; the finitely many exceptional configurations (Zsigmondy exceptions, small-field maximality exceptions, the outer automorphisms of A_6 , and so on) all have order at most 1.05×10^7 apart from two, $\text{PSL}(6, 2)$ and $\Omega^+(8, 2)$, which are handled in-text with substitute primes. In the interval $5 \times 10^5 \leq |S| \leq 1.05 \times 10^7$, the 38 groups $\text{PSL}(2, q)$ are deliberately routed to the exact uniform proof of Theorem 6.1; the other 13 groups, all smaller groups, and every sporadic group are settled by machine certificates (§5).

Three structural facts keep the family analysis manageable. First, when the two exhibited classes consist of *maximal* subgroups of S , all hypotheses of the machinery are automatic and only stability and the valuation inequality need checking. Second, for each twisted family used below, the automorphism structure induces no nontrivial permutation of the types of the relevant *relative* (twisted) diagram; field and diagonal automorphisms preserve those types. Thus the selected parabolic classes are stable under every automorphism. The twisted families, including the triality groups ${}^3D_4(q)$ and the Ree groups, are consequently among the easiest cases. Third, when a graph automorphism does fuse the relevant parabolic classes, the corresponding *flag* parabolics (intersections of fused maximal parabolics) become poset-maximal among stable classes, and a structural lemma (Lemma 3.6) shows they satisfy all needed hypotheses; this is the wreath-product analogue of the “novelty” phenomenon for maximal subgroups of almost simple groups.

Computer verification. The machine-certified ingredients were produced with GAP 4.16.0 [9], using its simple-group iterators, the character table library [2] and ATLAS data [6]. The environment pins CTblLib 1.3.11 and AtlasRep 2.1.11 and overlays the exact `ContainedConjugates` fix from GAP commit `b12f8342d641075d58fcbe62cc00dd433d7b8e18`; unpatched GAP 4.16.0 is rejected. Both GAP global pseudorandom sources are reset at process start to the recorded seed 1034, so internal library choices are reproducible; proof scripts contain no explicit random witness search. Proof-path scripts are deterministic and fail closed: any skipped case, soft failure, ambiguous identification, missing datum, version mismatch, or byte-level certificate difference is fatal to verification, and superseded exploratory scripts (including an earlier randomized witness search) are excluded from the certificate path. Every finite certificate records exact class positions and guard fingerprints as well as orders, the obstruction prime, and valuations. The scripts and logs, together with an arithmetic regression suite for the family proofs of §6 (7892 parameter instances; regression tests of the encoded formulas, not proofs of the universally quantified statements), are archived in a versioned release [24]. §5 specifies exactly what is claimed from the machine and how it is certified.

Selected structural steps are additionally formalized in Lean 4 and checked by its kernel: quotient inheritance of property P; the minimal-counterexample core (uniqueness of the minimal normal subgroup, non-solubility, soluble quotient, trivial centralizer); the product-supplement intersection, exact order, non-conjugacy, and maximality statements, under the normalized coordinate-action hypotheses of Convention 2.4; and the divisibility bridge and universal-in- k arithmetic contradiction. The formalization does not yet construct the decomposition $N \cong S^k$ from the structural input, the transitive wreath-product embedding and coordinate normalization, or the universal infinite-family arithmetic with its exception routing; these remain paper proofs. The exact formal coverage manifest is part of [24]. Generative AI was used in the research and preparation of this paper; a full declaration, specifying what was AI-assisted and how it was checked, appears before the bibliography.

Proof and verification roadmap. The proof has five separately auditable layers; each row of Table 1 states where the claim is proved and how it is verified.

Claim	Where	Verification
Minimal counterexample has unique minimal normal subgroup S^k , embeds in $X \wr S_k$	Prop. 2.3, Conv. 2.4	Structural proof (elementary); quotient inheritance and the minimal-order core (uniqueness, soluble quotient, trivial centralizer) kernel-checked in Lean; the S^k decomposition and wreath normalization are paper proofs, not yet formalized
Normalizers $B_V = N_G(V^k)$ of stable classes are maximal, non-conjugate, of exact order $t V ^k$	Lemmas 3.3–3.5, 3.6	Structural proof; exact normalizer intersection, supplement, order, maximality, and non-conjugacy kernel-checked in Lean under the normalized coordinate-action hypotheses of Conv. 2.4; parabolic input from [5]
Divisibility criterion: one stable pair with $d > v_p(x)$ excludes S^k for all $k \geq 2$	Thm. 4.1	Structural proof; subgroup-product/lower-divisibility bridge (conditional on the supplement conclusions) and the universal-in- k numerical contradiction kernel-checked in Lean
Stable pairs exist for every infinite family	§6, Thms. 6.1–6.6	Structural proofs; valuation inequalities re-checked mechanically over 7892 parameter instances (receipts in [24])
Sporadic groups, Tits group, and the finite base are excluded	Props. 5.1, 5.2	GAP certificates for every machine-routed case; the 38 upper-range $\text{PSL}(2, q)$ entries route to Theorem 6.1; scripts, logs and coverage cross-checks in [24]

TABLE 1. Proof and verification roadmap.

Every machine-backed claim in §5 names its script and committed log file; the repository README maps each paper claim to its certificate and gives a single command re-running the quick verification suite.

2. REDUCTION TO SOCLE ANALYSIS

Lemma 2.1 (Conjugation lemma). *If $G = AB$ for subgroups A, B , then $G = A^x B^y$ for all $x, y \in G$.*

Proof. Since $G = AB$, write $xy^{-1} = ab$ with $a \in A, b \in B$. Then, as subsets of G ,

$$A^x B^y = x^{-1} A x y^{-1} B y = x^{-1} A (ab) B y = x^{-1} (AB) y = x^{-1} G y = G. \quad \square$$

Consequently, property P depends only on the conjugacy classes of maximal subgroups: it says every pair of *distinct classes* has one (hence every) representative pair factorizing G .

Lemma 2.2. *Property P is inherited by quotients.*

Proof. Maximal subgroups of G/K are the images of the maximal subgroups of G containing K ; non-conjugate ones lift to non-conjugate ones, and factorizations project. $\square$

Proposition 2.3 (Minimal counterexample structure). *Let G be a non-soluble group with property P of least order. Then G has a unique minimal normal subgroup $N = S^k$ with S non-abelian simple and $k \geq 1$, $C_G(N) = 1$, G/N is soluble, and G embeds in $\text{Aut}(S) \wr S_k$ with N mapping onto $\text{Inn}(S)^k$ and G acting transitively on the k coordinates.*

Proof. Every proper quotient of G has property P (Lemma 2.2), hence is soluble by minimality of $|G|$. The soluble radical R of G is trivial: otherwise G/R is non-soluble (as R is soluble) with property P, contradicting minimality. So the socle of G is a direct product of non-abelian minimal normal subgroups. If $N_1 \neq N_2$ were two of them, then G/N_1 is soluble while $N_2 \cong N_2 N_1 / N_1 \leq G/N_1$ is non-soluble, a contradiction. Hence there is a unique minimal normal subgroup $N = S^k$, S non-abelian simple. Now $C_G(N) \cap N = Z(N) = 1$ since N is a direct product of non-abelian simple groups; if $C_G(N)$ were nontrivial it would contain a minimal normal subgroup of G , necessarily N by uniqueness, contradicting $C_G(N) \cap N = 1$. So $C_G(N) = 1$, and the conjugation action embeds G into $\text{Aut}(N) = \text{Aut}(S) \wr S_k$ in the standard way; minimal normality of N forces transitivity on coordinates. Exact published versions of the two standard structural inputs used here are [31, Lemma 2.10, pp. 173–174] for the automorphism group of a direct power and [19, p. 432] for the resulting transitive wreath realization in the unique-minimal-normal-subgroup setting. Finally G/N is soluble because it is a proper quotient. $\square$

Convention 2.4. For G as in Proposition 2.3 let G_1 be the stabilizer of the first coordinate and $\pi_1: G_1 \rightarrow \text{Aut}(S)$ the projection. Set $X := \pi_1(G_1) \cdot \text{Inn}(S)$, the *coordinate closure*; by coordinate-transitivity all coordinates give the same X up to conjugacy, and after conjugating inside $\text{Aut}(S) \wr S_k$ we may and do assume $G \leq X \wr S_k$ with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$. (For the normalization: fix $g_j \in G$ carrying coordinate 1 to coordinate j , with $g_1 = 1$, and let $a_j \in \text{Aut}(S)$ be its $1 \rightarrow j$ component. If $g \in G$ carries coordinate i to coordinate j through the component $c \in \text{Aut}(S)$, then $g_i g g_j^{-1}$ stabilizes coordinate 1 with component $a_i c a_j^{-1}$, which therefore lies in $\pi_1(G_1) \leq X$; so replacing G by $\delta G \delta^{-1}$, where $\delta := (a_1, \dots, a_k) \in \text{Aut}(S)^k$ (equivalently, conjugating by $(a_1^{-1}, \dots, a_k^{-1})$ in the exponent convention of Lemma 2.1), places every component of every element of G in X , and $a_1 = 1$ leaves X itself unchanged.)

For *coordinate* calculations we fix the following convention. Write $c_g(z) := g z g^{-1}$ (left conjugation), let S_i be the i -th coordinate factor of $N \cong S^k$, and let $\sigma_g \in S_k$ be the permutation determined by $c_g(S_j) = S_{\sigma_g(j)}$. There are component automorphisms $a_{g,i} \in \text{Aut}(S)$ such that

$$(c_g(z))_i = a_{g,i}(z_{\sigma_g^{-1}(i)}) \quad (z \in N, 1 \leq i \leq k):$$

the value arriving at coordinate i comes from the *inverse-permuted* source coordinate $\sigma_g^{-1}(i)$. From $c_{gh} = c_g \circ c_h$ these data compose by

$$\sigma_{gh} = \sigma_g \sigma_h, \quad a_{gh,i} = a_{g,i} a_{h,\sigma_g^{-1}(i)}.$$

After the normalization above, every $a_{g,i}$ lies in X . We retain the exponent convention $A^g = g^{-1}Ag$ of Lemma 2.1 for subgroup conjugation, so $A^g = c_{g^{-1}}(A)$; the two conventions are never interchanged. Write

$$x := |X/\text{Inn}(S)|, \quad t := |G/N|.$$

Then t divides $x^k k!$. Since property P and all criteria below are invariant under replacing G by a conjugate under $(a, \dots, a) \in \text{Aut}(S)^k$, statements may be proved after normalizing X up to $\text{Aut}(S)$ -conjugacy.

The following observation is not needed for the proof of Theorem 1.1 but explains why the pairs used later are of “supplement” type.

Remark 2.5 (Top-supplement pairs always factorize). If $M \supseteq N$ is maximal and B is maximal with $B \not\supseteq N$, then $BN = G$ (BN is a subgroup strictly containing the maximal B), and since $N \leq M$ we get $MB \supseteq NB = G$. Hence property P for G is equivalent to: P(G/N) together with the factorization of every pair of non-conjugate maximal subgroups not containing N .

3. PRODUCT SUPPLEMENTS

Fix G as in Convention 2.4: $N = S^k \leq G \leq X \wr S_k$, $t = |G/N|$, $x = |X/\text{Inn}(S)|$. For $V \leq S$ write $[V]$ for its S -conjugacy class. An S -class is X -stable if X permutes it, i.e. V^a is S -conjugate to V for every $a \in X$ (Inn-stability being automatic).

Definition 3.1. $\mathcal{P}_X(S) := \{[V] : 1 < V < S, N_S(V) = V, [V] \text{ } X\text{-stable}\}$, partially ordered by containment up to S -conjugacy ($[V] \leq [U]$ iff $V \leq U^s$ for some $s \in S$). Call V *normally saturating* if $\langle V^W \rangle = W$ for every overgroup $V \leq W \leq S$.

Remark 3.2. If V is a *maximal* subgroup of S then, whenever $[V]$ is X -stable: $[V] \in \mathcal{P}_X(S)$ (maximal subgroups of a non-abelian simple group are self-normalizing), $[V]$ is a maximal element of the poset, and V is normally saturating (its only overgroups are V and S , and $\langle V^S \rangle = S$ by simplicity). So for maximal classes the lemmas below apply with no further hypotheses.

Lemma 3.3 (Existence). *Let $[V] \in \mathcal{P}_X(S)$ and set $B_V := N_G(V^k)$. Then*

- (1) $B_V \cap N = N_S(V)^k = V^k$;
- (2) $B_V N = G$;
- (3) $|B_V| = t \cdot |V|^k$ *exactly*.

Proof. (1) An element of $N = S^k$ normalizes V^k iff each coordinate normalizes V . (2) Frattini-type argument: let $g \in G$. In the coordinate-action notation of Convention 2.4, $(V^k)^g = c_{g^{-1}}(V^k)$, and since every coordinate of V^k is V ,

$$(V^k)^g = \prod_{i=1}^k a_{g^{-1},i}(V),$$

each factor the image of V under a component automorphism $a_{g^{-1},i} \in X$. By X -stability there are $s_i \in S$ with $a_{g^{-1},i}(V) = V^{s_i}$. So $(V^k)^g$ is N -conjugate to V^k : there is $n \in N$ with $(V^k)^{gn} = V^k$, i.e. $gn \in B_V$, so $g \in B_V N$. (3) $|B_V| = |B_V N| \cdot |B_V \cap N|/|N| = t |V|^k$. $\square$

Lemma 3.4 (Maximality). *If in addition $[V]$ is a maximal element of $\mathcal{P}_X(S)$ and V is normally saturating, then B_V is maximal in G .*

Proof. Let $B_V \leq H \leq G$. Since $H \supseteq B_V$ and $B_V N = G$ we have $HN = G$, so H surjects onto G/N and in particular acts coordinate-transitively.

Step 1 (H 's twists recover X). Let H_1, G_1 be the coordinate-1 stabilizers. For $g \in G_1$ write $g = hn$ ($h \in H$, $n \in N$); n fixes every coordinate, hence so does h , so $h \in H_1$ and

$\pi_1(G_1) = \pi_1(H_1) \cdot \pi_1(N \cap G_1) = \pi_1(H_1) \cdot \text{Inn}(S)$. Therefore $\pi_1(H_1)$ and $\text{Inn}(S)$ together generate X , and a class of subgroups of S is X -stable iff it is stable under $\pi_1(H_1)$.

Step 2 (Goursat + saturation: $H \cap N$ is a product). Let $T := H \cap N \supseteq V^k$, let $W_i := \pi_i(T) \leq S$ be the coordinate projections and $K_i := T \cap S_i$ the coordinate kernels (S_i the i -th factor). Then $V \leq K_i$ (as $V^k \cap S_i = V$ in coordinate i), $K_i \trianglelefteq W_i$, and $V \leq W_i$. Since K_i is a normal subgroup of W_i containing V , it contains $\langle V^{W_i} \rangle = W_i$ by saturation. So $K_i = W_i$ for every i and $T = \prod_i W_i$.

Step 3 (the W_i force $H = B_V$ or $H = G$). T is H -invariant. For $h \in H_1$, projecting $T^h = T$ to the first coordinate gives $W_1^{\pi_1(h)} = W_1$, so $[W_1]$ is $\pi_1(H_1)$ -stable, hence X -stable by Step 1. For any i , coordinate-transitivity of H supplies $h \in H$ carrying coordinate i to coordinate 1 through a component $c \in X$, and projecting $T^h = T$ gives $W_1 = W_i^c$; applying X -stability of $[W_1]$ to c^{-1} gives $[W_i] = [W_1]$. Write $[W]$ for this common X -stable class. If $W = S$ then $T = N$, so $H \supseteq N$ and $H = HN = G$. Otherwise $1 < V \leq W < S$. Form the normalizer tower $W \leq N_S(W) \leq N_S(N_S(W)) \leq \dots$; it is strictly increasing until it becomes self-normalizing, and it cannot reach S (the penultimate term would be a proper normal subgroup of the simple group S). Its terminal member U is proper, self-normalizing, and $[U]$ is X -stable ($N_S(W^a) = N_S(W)^a$, so stability propagates up the tower), i.e. $[U] \in \mathcal{P}_X(S)$ with $[U] \geq [W] \geq [V]$. Maximality of $[V]$ in $\mathcal{P}_X(S)$ forces $[U] = [V]$; then $|U| = |V|$ and $V \leq W \leq U$ gives $V = W = U$. So $T = V^k$, hence $|H| = |HN| \cdot |H \cap N|/|N| = t|V|^k = |B_V|$ and $H = B_V$. $\square$

Lemma 3.5 (Non-conjugacy). *Distinct $[V] \neq [W]$ in $\mathcal{P}_X(S)$ give non-conjugate B_V, B_W .*

Proof. If $B_V^g = B_W$ then $(B_V \cap N)^g = B_W \cap N$ (as $N \trianglelefteq G$), i.e. $(V^k)^g = W^k$. Reading off any coordinate, W is the image of V under an element of X composed with a coordinate permutation, so $[W] = [V^a] = [V]$ for some $a \in X$ by X -stability, a contradiction. $\square$

The next lemma supplies, structurally, all hypotheses of Lemma 3.4 for the non-maximal (“novelty”) classes used in §6: the standard parabolic subgroups whose maximal parabolic overgroups are fused by a graph automorphism. We use the notation of [5]: S an untwisted simple group of Lie type with split BN -pair of rank $\ell \geq 2$ (all our applications are untwisted), B a Borel subgroup, Δ the node set of the Dynkin diagram, and for $J \subseteq \Delta$, $Q_J \supseteq B$ the standard parabolic whose Levi subgroup L_J is generated by the maximal torus T and the root subgroups of J ; the maximal parabolic omitting node i is $P_i = Q_{\Delta \setminus \{i\}}$.

Lemma 3.6 (Parabolic novelties). *Fix $J \subseteq \Delta$ and put $V := Q_J$. Suppose X is such that*

- (a) *the S -class $[Q_J]$ is X -stable, and*
- (b) *for every $J \subsetneq K \subsetneq \Delta$ the class $[Q_K]$ is not X -stable.*

Then $[V] \in \mathcal{P}_X(S)$, $[V]$ is a maximal element of $\mathcal{P}_X(S)$, and V is normally saturating. Consequently Lemmas 3.3–3.5 apply to V with no computational hypothesis.

Proof. Parabolic subgroups are self-normalizing [5, Thm. 8.3.3, p. 112], so $[V] \in \mathcal{P}_X(S)$ by (a).

Overgroups. Every subgroup containing V contains B , and every subgroup containing a Borel subgroup is a parabolic Q_K , $K \supseteq J$, of the same BN -pair [5, Thm. 8.3.2, p. 112]; moreover Q_K and $Q_{K'}$ are S -conjugate iff $K = K'$ [5, Thm. 8.3.3, p. 112].

Poset-maximality. If $[U] \in \mathcal{P}_X(S)$ with $V \leq U^s < S$ for some s , then $U^s = Q_K$ for some $J \subseteq K \subsetneq \Delta$; $K = J$ gives $[U] = [V]$, while $J \subsetneq K$ contradicts (b), since $[U] = [Q_K]$ would be X -stable.

Saturation. Let $V \leq W \leq S$, so $W = Q_K$ ($K \supseteq J$) or $W = S$; we must show $\langle V^W \rangle = W$. The normal closure of V in W contains that of B . Write $W = U_K \rtimes L_K$ (Levi decomposition; for $W = S$ read $U_K = 1$, $L_K = S$). Then $U_K \leq B$, and modulo U_K the image of B is a Borel subgroup $B_L = T \cdot (U \cap L_K)$ of L_K containing the full torus T . For each node $\alpha \in K$

the reflection representative n_α lies in L_K , and $U_{-\alpha} = U_\alpha^{n_\alpha}$. The subgroup $\langle B_L^{L_K} \rangle$ contains every L_K -conjugate of every subgroup of B_L ; in particular it contains T , each U_α ($\alpha \in K$ positive), and each $U_{-\alpha} = U_\alpha^{n_\alpha}$. These generate L_K by the root-subgroup description of the Levi decomposition [5, §8.5, pp. 118–120]. So $\langle B^W \rangle \supseteq U_K \cdot L_K = W$. $\square$

4. THE DIVISIBILITY CRITERION

For a prime p , v_p denotes the p -adic valuation and $s_p(k)$ the digit sum of k in base p , so that $v_p(k!) = (k - s_p(k))/(p - 1)$.

Theorem 4.1 (Divisibility criterion). *Let S, X be as in Convention 2.4 and let $[V] \neq [W]$ be maximal elements of $\mathcal{P}_X(S)$, both normally saturating. If some prime p satisfies*

$$d := v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x),$$

then no group G with unique minimal normal subgroup S^k and coordinate closure X , for any $k \geq 2$, has property P.

Proof. By Lemmas 3.3–3.5, B_V and B_W are non-conjugate maximal subgroups of G , so property P forces $G = B_V B_W$ and

$$|B_V \cap B_W| = \frac{|B_V| |B_W|}{|G|} = t \cdot \left(\frac{|V| |W|}{|S|} \right)^k \in \mathbb{Z},$$

whence $p^{dk} \mid t$. But

$$v_p(t) \leq k v_p(x) + v_p(k!) = k v_p(x) + \frac{k - s_p(k)}{p - 1} < k(v_p(x) + 1) \leq dk,$$

a contradiction. $\square$

Remark 4.2. (i) The criterion is independent of k : this is what closes infinite families, which no counting bound can do, since $(k!)^{1/k} \rightarrow \infty$ outgrows any fixed ratio $|V||W|/|S|$. (ii) Only the *existence* of B_V, B_W is used; no classification of the maximal subgroups of G (or even of S) is needed. In particular the criterion is insensitive to completeness of tabulated maximal-subgroup data: it needs only that the two exhibited subgroups are maximal (or Lemma 3.6-novelties) and their classes stable.

Theorem 4.3 ($k = 1$: the almost simple case). *The same criterion excludes the almost simple case: for $S \leq G \leq \text{Aut}(S)$ put $X := G$, $x := |G/S| = t$. For $[V]$ a maximal element of $\mathcal{P}_X(S)$, normally saturating, $B_V := N_G(V)$ satisfies $B_V \cap S = V$, $B_V S = G$, $|B_V| = t|V|$, and B_V is maximal in G . Hence if $[V] \neq [W]$ are two such classes and $d = v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x)$ for some prime p , then G does not have property P.*

Proof. $B_V S = G$ is the Frattini argument via X -stability as in Lemma 3.3; $B_V \cap S = N_S(V) = V$; the maximality of B_V is the proof of Lemma 3.4 without the Goursat step (normal saturation enters only through that step and is not needed at $k = 1$; the hypothesis is kept for uniformity with Theorem 4.1): $T := H \cap S$ is an H -invariant subgroup containing V ; if $T = S$ then $H \supseteq S$ and $H = HS = G$; otherwise the normalizer tower of T ends at a proper self-normalizing subgroup whose class is X -stable (here $G = B_V S \leq HS$ forces $HS = G$, so $X = G$ is generated by the images of H and S , and stability may be tested against the H -invariance of T , exactly as in Step 1 of Lemma 3.4) and lies above $[V]$, so poset-maximality forces $T = V$ and $H \leq N_G(V) = B_V$. The two normalizers are non-conjugate: if $B_V^g = B_W$ with $g \in G$, then $V^g = (B_V \cap S)^g = B_W \cap S = W$, and X -stability of $[V]$ (here $X = G$) gives $[W] = [V]$, a contradiction. Property P then forces $G = B_V B_W$, so $t |V||W|/|S| \in \mathbb{Z}$, i.e. $p^d \mid t = x$, contradicting $d > v_p(x)$. $\square$

Convention 4.4. From now on, “ S is excluded” means: for every X with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$ there exist classes as in Theorem 4.1; consequently no finite group G with unique minimal normal subgroup S^k ($k \geq 2$, any admissible embedding) has property P. The almost simple branch $k = 1$ is supplied separately by Tikhonenko–Tyutyaynov [28].

5. THE MACHINE BASE

Proposition 5.1. *Every non-abelian simple group S with $|S| \leq 1.05 \times 10^7$ is excluded (in the sense of Convention 4.4).*

The proof of Proposition 5.1 combines the uniform $\text{PSL}(2, q)$ argument of Theorem 6.1 with a finite computation in GAP 4.16.0 [9], organized as follows; every machine claim is backed by a committed log file (each sweep named below corresponds to a script `sweep<name>.g` and a certificate `sweep<name>.log` in [24], whose README maps every paper claim to its scripts and certificates), and the group inventory is cross-checked against GAP’s `SimpleGroupsIterator` in two installments, so that no group in range is skipped: the 47 groups with $|S| < 5 \times 10^5$ (receipt `verify_coverage.log`) and the 51 groups with $5 \times 10^5 \leq |S| \leq 1.05 \times 10^7$ (receipt `verify_coverage_big.log`). Of the latter 51, the 38 $\text{PSL}(2, q)$ entries terminate in Theorem 6.1 and its arithmetic receipt rather than a redundant GAP maximal-subgroup enumeration; sweeps J3 and J6 exhaust the other 13. The receipt also re-checks that the only two survivors among those 13, $\text{Sp}(4, 4)$ at $x = 4$ and $\text{PSL}(5, 2)$ at $x = 2$, are closed by the novelty certificates of item (2).

- (1) *Maximal-class pairs (sweeps J, J2, J4, J3, J5, J6).* For each machine-routed simple S in range and each X with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$, the program lists the X -stable classes of maximal subgroups of S , and searches pairs $([V], [W])$ and primes p satisfying the inequality of Theorem 4.1. For each conjugacy class of subgroups of $\text{Out}(S)$, the frozen program audits every quotient-class representative and emits the lexicographically canonical mathematical receipt; disagreement inside one quotient class is fatal. Raw GAP class positions and pc-generator words are not identifiers: each **XCASE** is ordered and named by a SHA-256 fingerprint of the complete induced permutation action on the pinned maximal-subgroup-class list, together with intrinsic quotient/kernel data; the rare collision is refined by the structure of the almost-simple preimage. The independent parser requires a contiguous, fingerprint-distinct **XCASE** inventory, including survivor cases, and binds every successful **CERT** to exactly one such record. For all but six socles in range this succeeds for every X . A logged certificate line records $(|V|, |W|, p, d)$; e.g. for $S = A_5$: $(|A_4|, |S_3|, 5, 1)$, excluding A_5^k for all k and all X , the first unbounded- k closure. In addition, the alternating groups A_{11} – A_{14} , whose orders exceed the base bound, carry certificates of the same form for every X (sweep J5, log `sweepJ5_smallAn.log`); Theorem 6.2 uses these for $11 \leq n \leq 14$.
- (2) *Novelty pairs (sweeps K, K2, K3, K4).* Six (S, X) families admit too few stable maximal classes, all for the expected reason (an outer automorphism fuses classes): $\text{PSL}(3, 2)$ and $\text{PSL}(2, 11)$ at $x = 2$; A_6 for $X/\text{Inn} \in \{2_2, 2_3, 2^2\}$; $\text{PSL}(3, 4)$ for several X ; $\text{Sp}(4, 4)$ at $x = 4$; $\text{PSL}(5, 2)$ at $x = 2$. Each is closed by a pair involving non-maximal classes that are maximal in $\mathcal{P}_X(S)$, with normal saturation verified computationally (sweep K2) or structurally (Lemma 3.6). Sweep K2 is embedding-aware. For a fixed representative V it enumerates actual overgroups with `IntermediateSubgroups(S, V)`; independently it enumerates every $N_S(V) \backslash S/W$ double coset whose conjugate of V lies in W ; and it compares both with `ContainedConjugates(S, W, V)` *without* the one-result flag, using the pinned, regression-tested implementation noted in the introduction. Any orbit mismatch or normal-closure mismatch is fatal: every embedding orbit of V in each overgroup

class is checked, not merely one representative per pair of subgroup classes. Every record names the S -class position, order, index, normalizer order, a SHA-256 canonical numeric class fingerprint, the overgroup class, embedding-orbit identifier, and normal-closure order. The K certificates are cross-linked to those exact fingerprints, and the independent checker rejects any selected class whose K2 status is not **SATURATES**. Thus same-order nonconjugate classes are never identified by order alone. Sweep K uses the same exact quotient-class scheme, with the induced action on the complete pinned list of S -subgroup classes. Sample certificates: $\text{PSL}(3, 2)$ at $x = 2$ via $(S_3, 7:3)$ at $p = 2$, $d = 2 > 1$; $\text{Sp}(4, 4)$ at $x = 4$ via (Borel, subfield $\text{Sp}(4, 2)$) at $p = 5$ and 17; $\text{PSL}(5, 2)$ at $x = 2$ via the incident-flag parabolics $(Q_{\{2,3\}}, Q_{\{1,4\}})$ (the point-hyperplane and line-3-space flag stabilizers, in the standard parabolic notation of Lemma 3.6) at $p = 5$ and 31. The last two are instances of the uniform constructions of Theorem 6.6 below.

Proposition 5.2. *Every sporadic simple group, and the Tits group ${}^2F_4(2)'$, is excluded.*

Proof. Six of the 26 sporadic groups ($M_{11}, M_{12}, M_{22}, M_{23}, J_1, J_2$) have order at most 1.05×10^7 and fall under Proposition 5.1. The remaining 20 ($M_{24}, J_3, J_4, HS, McL, Co_1, Co_2, Co_3, Suz, He, Ru, O'N, Fi_{22}, Fi_{23}, Fi'_{24}, HN, Ly, Th, B, M$) and the Tits group are handled by a dedicated computation (sweep M; script `sweepM_sporadic.g`, certificate `sweepM_sporadic.log`): stable maximal classes and pair searches use the ATLAS maximal-subgroup data via the GAP character table library [2, 6], pinned to CTblLib 1.3.11 and AtlasRep 2.1.11. Every record gives the exact character-table identifier and **Maxes** list position. If $\text{Out}(S) = 1$, stability is automatic. If $|\text{Out}(S)| = 2$, the program accepts a class only when its order occurs exactly once in the complete pinned **Maxes** list; automorphisms preserve order, so that exact class is stable. No stored class fusion is claimed or used. Each of the resulting 42 selected class records is also bound, by exact group, list position, identifier, and order, to a published maximal-subgroup record. For the non-Monster sporadics these are Wilson's corrected tables [30, §4, pp. 66–68], whose published-proof provenance is summarized in [3, §2, pp. 23–24]; the Tits entries are covered independently by [29, pp. 553–563] and [27]; and the current Monster entries are Theorems 1.1–1.2 and Table 1 of [8, pp. 862–863]. Sample certificates: the Monster via positions 45, 46, (59:29, 41:40) at $p = 71$; the Baby Monster via positions 29, 30, $(L_2(11).2, 47:23)$ at $p = 31$. By the remark after Theorem 4.1, completeness is logically needed here only for the unique-order stability argument when $|\text{Out}(S)| = 2$. The exact source map is machine-checked against the certificate rather than maintained only in prose. $\square$

Beyond the base, the family proofs of §6 carry their own numeric receipts: the valuation inequalities of Theorems 6.1, 6.2, and 6.4–6.6 were verified mechanically over large parameter ranges (1272 prime powers for Theorem 6.1; $n \leq 10^4$ for Theorem 6.2; 7892 parameter instances, ranks up to 25 and fields up to 3000, for Theorems 6.4–6.6), with zero failures and with the Zsigmondy-exception set exactly as documented in the proofs; the Levi-exponent inequalities behind Theorem 6.3 are checked separately by the independent symbolic verifier in [24]. These finite-range runs are regression checks, not proofs of the statements quantified over all ranks and fields.

6. THE INFINITE FAMILIES

Throughout this section S is simple, X is any group with $\text{Inn}(S) \leq X \leq \text{Aut}(S)$, and $x = |X/\text{Inn}(S)|$. By Remark 3.2, when the two exhibited classes consist of maximal subgroups of S , only their X -stability and the valuation inequality require proof. We use Zsigmondy's theorem [33] throughout: for integers $a \geq 2$, $e \geq 3$ with $(a, e) \neq (2, 6)$, there is a prime r with $\text{ord}_r(a) = e$ (a *primitive prime divisor* of $a^e - 1$); such r satisfies $r \equiv 1 \pmod{e}$. For $q = p^f$

we write $\Phi_e = \Phi_e(q)$ for cyclotomic values and always apply Zsigmondy to the prime p with exponent ef , so $\text{ord}_r(p) = ef$ and $r \equiv 1 \pmod{ef}$.

6.1. Projective special linear groups of rank one.

Theorem 6.1. $S = \text{PSL}(2, q)$, $q = p^f \geq 4$, is excluded.

Proof. For $q \in \{4, 5, 7, 8, 9, 11\}$ this is machine-certified by the explicit lower-range J/K certificates of §5 (not by an invocation of Proposition 5.1). All other simple parameters are covered uniformly below: odd $q \geq 13$ by Case A and even $q = 2^f$, $f \geq 3$, by Case B (including $q = 256$, without invoking the finite base). Recall $|S| = q(q^2 - 1)/e$ with $e = \gcd(2, q - 1)$, and $\text{Out}(S) = C_e \times C_f$, so $x \mid ef$: that is, $x \mid 2f$ for q odd and $x \mid f$ for q even.

Case A: q odd, $q \geq 13$. Let $U := N_S(T_+)$ and $V := N_S(T_-)$ be the normalizers of the split and nonsplit maximal tori: dihedral of orders $q - 1$ and $q + 1$, and maximal in S for $q \geq 13$ by Dickson's theorem [7, 14]; see also [1, Tables 8.1–8.2, p. 376, and Table 8.7, p. 380]. All split (resp. nonsplit) maximal tori are S -conjugate and every automorphism maps one to one (there is no graph automorphism in type A_1 ; diagonal and field automorphisms act algebraically), so $[U]$ and $[V]$ are X -stable for every X . Now $|U||V|/|S| = e/q$, so for the defining prime p : $d = v_p(|S|) - 0 - 0 = f$, while $x \mid 2f$ gives $v_p(x) \leq v_p(f) \leq \log_p f < f = d$. Theorem 4.1 applies.

Case B: $q = 2^f$, $f \geq 3$. Here $|S| = q(q^2 - 1)$ and $x \mid f$. Let U be a Borel subgroup (the normalizer of a Sylow 2-subgroup), of order $q(q - 1)$, and $V := N_S(T_+)$, dihedral of order $2(q - 1)$, maximal for $q \geq 8$ [7, 14]; see also [1, Tables 8.1–8.2, p. 376]. $[U]$ is the class of Sylow-2 normalizers and $[V]$ the unique class of split torus normalizers: both X -stable. Now $|U||V|/|S| = 2(q - 1)/(q + 1)$ with $q + 1$ odd and coprime to $2(q - 1)$, so it suffices to find a prime $r \mid q + 1$ with $v_r(q + 1) > v_r(f)$. If $f \neq 3$, take r a primitive prime divisor of $2^{2f} - 1$: then $r \mid (2^{2f} - 1)/(2^f - 1) = q + 1$ and $r \equiv 1 \pmod{2f}$, so $r > f$ and $v_r(x) = 0 < v_r(q + 1) = d$. If $f = 3$ ($q = 8$): $q + 1 = 9$, $r = 3$: $d = 2 > 1 \geq v_3(x)$. $\square$

6.2. Alternating groups.

Theorem 6.2. $S = A_n$, $n \geq 5$, is excluded.

Proof. For $n \leq 14$ this is machine-certified: $n \leq 10$ falls under Proposition 5.1 (including $n = 6$, whose exceptional outer automorphisms fuse classes; all five X are handled there), and $11 \leq n \leq 14$ by the sweep-J5 certificates of §5. Assume $n \geq 15$, so $\text{Aut}(A_n) = S_n$ and $x \mid 2$.

Let p be the largest prime $\leq n$. For $m < n/2$ let $M_m := (S_m \times S_{n-m}) \cap A_n$, and for n even let $I := (S_{n/2} \wr S_2) \cap A_n$. Each M_m is maximal in A_n , and I is maximal outside a finite list of exceptions all with $n \leq 14$ [17, Thm. 1 and Tables I–II, pp. 366–368]. Conjugation by S_n preserves the partition shape (resp. block structure), each shape is a single A_n -class, and $\text{Aut}(A_n) = S_n$ for $n \neq 6$: all these classes are X -stable for every X .

For any two distinct classes among $\{M_m : n - p < m < n/2\} \cup \{I \text{ if } n \text{ even and } n/2 < p\}$: $v_p(|A_n|) = 1$ (Bertrand: $n/2 < p \leq n$), while v_p of each subgroup order is 0 (all parts of the partitions involved are $< p$), and $v_p(x) \leq v_p(2) = 0$. So $d = 1 > 0$ and Theorem 4.1 applies, provided two such classes exist. The number of admissible intransitive m is $p - \lfloor n/2 \rfloor - 1$, plus one for I when n is even; two classes exist whenever $p \geq n/2 + 3$. By Nagura's theorem [22] (a prime in $(x, 6x/5]$ for $x \geq 25$), for $n \geq 31$ there is a prime $p \in (5n/6, n]$, and $5n/6 > n/2 + 3$ for $n > 18$. For $15 \leq n \leq 30$ the largest prime $\leq n$ satisfies the bound by inspection ($p = 13, 13, 17, 17, 19, 19, 19, 19, 23, 23, 23, 23, 23, 29, 29$ for $n = 15, \dots, 30$). $\square$

6.3. Lie type without graph symmetry.

Theorem 6.3. *Let S be a simple group of Lie type of rank ≥ 2 over $\mathbb{F}_q$, $q = p^f$, of a type with no symmetry of its Dynkin diagram in the given characteristic:*

$$C_n = \mathrm{PSp}(2n, q) \ (n \geq 2, \ p \text{ odd if } n = 2), \quad B_n = \Omega(2n+1, q) \ (n \geq 3, \ q \text{ odd}), \\ G_2(q) \ (p \neq 3), \quad F_4(q) \ (p \neq 2), \quad E_7(q), \quad E_8(q).$$

Then S is excluded.

Proof. Write h for the relevant torus exponent: $h = 2n$ for B_n/C_n , and $h = 6, 12, 18, 30$ for G_2, F_4, E_7, E_8 .

Let P_1, P_2 be maximal parabolic subgroups from two distinct classes (rank ≥ 2 guarantees these; take the ends of the diagram). They are maximal subgroups of S : any overgroup of a parabolic contains a Borel subgroup, and every subgroup containing a Borel is parabolic [5, Thm. 8.3.2, p. 112].

Stability. $\mathrm{Aut}(S)$ is generated by inner, diagonal, field and graph automorphisms [25], [4, Def. 3.1 and Thm. 3.4, pp. 31–32]. Diagonal and field automorphisms preserve types of vertices of the building, hence each parabolic class; type-permuting automorphisms arise only from diagram symmetries, absent for the listed types in the listed characteristics (the exceptional symmetries of B_2/C_2 , F_4 in characteristic 2 and G_2 in characteristic 3 are excluded by hypothesis and treated in Theorem 6.6). So every parabolic class is X -stable for every X .

The obstruction prime. Let r be a primitive prime divisor of $p^{hf} - 1$. Then: (i) $r \mid |S|$, since the order formula of each listed type contains the factor $q^h - 1$ (equivalently $\Phi_h(q)$) [5, §14.1]. (ii) r divides no maximal parabolic order: $|P_i| = q^{N_i} |L_i| (q-1)^{c_i}$ with L_i the semisimple part of the Levi factor, whose order is a product of a q -power and factors $q^j - 1$ with $j \leq h_i$, h_i the largest torus exponent of L_i ; in every listed case every proper Levi has $h_i < h$ (B_n/C_n : Levi types $A_{i-1} \times C_{n-i}$ give $j \leq \max(i, 2(n-i)) < 2n$; G_2 : Levis A_1 , $j \leq 2$; F_4 : Levis $B_3, C_3, A_2 \times A_1$, $j \leq 6$; E_7 : Levis $E_6, D_6, A_6, \dots$, exponents ≤ 12 ; E_8 : Levis E_7, D_7, A_7 , exponents ≤ 18). By primitivity r divides none of these, nor q , nor $q-1$. (iii) $r \nmid x$: $r \equiv 1 \pmod{hf}$ with $h \geq 4$, so $r > hf \geq 4f \geq d_S f \geq x$, where $d_S \leq \gcd(2, q-1)^2 \leq 4$ bounds the diagonal part and there is no graph part.

So $d = v_r(|S|) \geq 1 > 0 = v_r(x)$ for the pair $([P_1], [P_2])$, and Theorem 4.1 applies, provided r exists.

Zsigmondy exceptions. $p^{hf} = 2^6$ forces $p = 2$ and $hf = 6$: either $G_2(2)$ (not simple; $G_2(2)' \cong \mathrm{PSU}(3, 3)$ is in the machine base) or $\mathrm{Sp}(6, 2)$ (machine base). For $h \in \{12, 18, 30\}$, $hf = 6$ is impossible; B_n has q odd. $\square$

6.4. Twisted groups of twisted rank at least two.

Theorem 6.4. *Let S be one of*

$$\mathrm{PSU}(n, q) \ (n \geq 4), \quad P\Omega^-(2n, q) = {}^2D_n(q) \ (n \geq 4), \\ {}^3D_4(q), \quad {}^2E_6(q), \quad {}^2F_4(q) \ (q = 2^f, \ f \text{ odd } \geq 3).$$

Then S is excluded.

Proof. These are exactly the twisted types of twisted BN -pair rank ≥ 2 , so S has at least two classes of maximal parabolic subgroups P_1, P_2 , maximal in S as in Theorem 6.3. The required applicability is explicit: every finite group of Lie type in defining characteristic has a split BN -pair [13, Example 1.16 and §1.3.5, pp. 10–11], and Carter's overgroup and self-normalizer theorems are stated for an arbitrary group with a BN -pair [5, Thms. 8.3.2–8.3.3, p. 112].

Stability. We claim every parabolic class is X -stable for every X . Every $a \in \mathrm{Aut}(S)$ permutes the parabolic subgroups: they are exactly the overgroups of the normalizers of the Sylow p -subgroups (Borel subgroups), an automorphism-invariant description; hence a permutes the

finitely many parabolic classes, preserving inclusion up to conjugacy, i.e. acts on the type set of the twisted building. Now use the automorphism description of [4, Def. 3.3 and Thm. 3.4, p. 32], which sets the separate graph factor equal to 1 for a twisted group. The inner-diagonal and Frobenius representatives centralizing the defining Steinberg endomorphism preserve each orbit of simple roots, hence each node of the relative (twisted) diagram. Inner automorphisms fix classes, while the standard outer representatives normalize the standard Borel and torus and therefore preserve the class of each relative type. Thus they map every standard parabolic Q_J to a conjugate of the same type. Hence every automorphism fixes every parabolic class. (For the rank-two group ${}^2F_4(q)$ one can also see directly that the two maximal parabolic classes cannot be interchanged: their Levi factors ${}^2B_2(q)$ and $\mathrm{SL}_2(q)$ are non-isomorphic. Note that a diagram-asymmetry argument alone would not suffice at twisted rank two: the twisted diagrams of $\mathrm{PSU}(4, q)$ and $\mathrm{PSU}(5, q)$ have a symmetric underlying graph, and the untwisted example $\mathrm{Sp}(4, 2^f)$ shows that a multiple bond by itself does not prevent a type-swapping automorphism.) So every parabolic class is X -stable for every X .

The obstruction prime. Choose the exponent e and r a primitive prime divisor of $p^e - 1$ per type; in each case r divides $|S|$ but neither $|P_1|$ nor $|P_2|$:

- $\mathrm{PSU}(n, q)$, n odd: $e = 2nf$, so $r \mid \Phi_{2n}(q) \mid q^n + 1$, which divides $|\mathrm{SU}(n, q)| = q^{n(n-1)/2} \prod_{i=2}^n (q^i - (-1)^i)$. The Levi factor of P_i ($i = 1, 2$) is a central product of $\mathrm{GL}_i(q^2)$ and $\mathrm{GU}(n - 2i, q)$: its p' -order is a product of factors $q^{2j} - 1$ ($j \leq 2$) and $q^j - (-1)^j$ ($j \leq n - 2$), all dividing $p^m - 1$ with $m \leq \max(4f, 2(n - 2)f) < 2nf$.
- $\mathrm{PSU}(n, q)$, n even: $e = 2(n - 1)f$, $r \mid \Phi_{2(n-1)}(q) \mid q^{n-1} + 1$ (the $i = n - 1$ factor). Levi exponents are $\leq \max(4f, 2(n - 3)f, (n - 2)f) < 2(n - 1)f$ for $n \geq 4$.
- ${}^2D_n(q)$, $n \geq 4$: $e = 2nf$, $r \mid \Phi_{2n}(q) \mid q^n + 1$, which divides $q^{n(n-1)}(q^n + 1) \prod_{i=1}^{n-1} (q^{2i} - 1)$. Levi of P_i : $\mathrm{GL}_i(q) \times O^-(2(n - i), q)$ -type, exponents $\leq \max(2f, 2(n - 1)f) < 2nf$.
- ${}^3D_4(q)$: $e = 12f$, $r \mid \Phi_{12}(q) = q^4 - q^2 + 1$, which divides $q^8 + q^4 + 1 = (q^4 + q^2 + 1)(q^4 - q^2 + 1)$, hence $|{}^3D_4(q)| = q^{12}(q^8 + q^4 + 1)(q^6 - 1)(q^2 - 1)$. The two Levi factors are $\mathrm{SL}_2(q^3) \cdot (q - 1)$ and $\mathrm{SL}_2(q) \cdot (q^3 - 1)$ (mod centers): p' -orders divide $(q^6 - 1)(q - 1)$ and $(q^2 - 1)(q^3 - 1)$, exponents $\leq 6f < 12f$.
- ${}^2E_6(q)$: $e = 18f$, $r \mid \Phi_{18}(q) \mid q^9 + 1$, which divides $q^{36}(q^{12} - 1)(q^9 + 1)(q^8 - 1)(q^6 - 1)(q^5 + 1)(q^2 - 1)$. The four maximal parabolic classes correspond to the four τ -orbits of nodes of E_6 (τ the order-two symmetry): $\{2\}$, $\{4\}$, $\{1, 6\}$, $\{3, 5\}$; the complementary τ -stable subdiagrams are A_5 , $A_2 \times A_1 \times A_2$, D_4 , $A_2 \times A_1 \times A_1$ with induced twists, so the Levi derived groups are $\mathrm{SU}(6, q)$, $\mathrm{SL}_2(q) \times \mathrm{SL}_3(q^2)$, ${}^2D_4(q)$ -type, and $\mathrm{SL}_3(q) \times \mathrm{SL}_2(q^2)$, with cyclotomic exponents $\leq 10f, 6f, 8f, 4f$ respectively, all $< 18f$. Any two classes serve as P_1, P_2 . This diagram derivation agrees with the published list (${}^2A_5(q)$, ${}^2D_4(q)$, $A_1(q) \times A_2(q^2)$, $A_1(q^2) \times A_2(q)$) in [21, Lemma 6.8, p. 82].
- ${}^2F_4(q)$, f odd ≥ 3 : $e = 12f$, $r \mid \Phi_{12}(q) \mid q^6 + 1$, which divides $|{}^2F_4(q)| = q^{12}(q^6 + 1)(q^4 - 1)(q^3 + 1)(q - 1)$. The two Levi derived groups are ${}^2B_2(q) \cong \mathrm{Sz}(q)$ and $\mathrm{SL}_2(q)$ [11, §2.9]: p' -orders divide $(q^2 + 1)(q - 1)^2$ and $(q^2 - 1)(q - 1)$, exponents $\leq 4f < 12f$. (${}^2F_4(2)$ is not simple; its derived group, the Tits group, is in the machine base.)

$r \nmid x$. In every case x divides $d_S \cdot cf$ where the diagonal order d_S divides $q + 1$ (types 2A , 2E_6) or 4 (type 2D) or is 1 (3D_4 , 2F_4), and the field part has order cf with $c \leq 3$. Since $r \equiv 1 \pmod{e}$ with $e \geq 6f$, r exceeds the field-part order; $r \mid d_S$ would give either $r \mid q + 1$, so $\mathrm{ord}_r(p) \mid 2f < e$, or $r \leq 4 < r$, both impossible. Hence $v_r(x) = 0 < 1 \leq d$ and Theorem 4.1 applies.

Zsigmondy exceptions. r fails to exist only if $p^e = 2^6$. Here $e \geq 6$ always, and $e = 6$ occurs only for $\mathrm{PSU}(4, q)$ with $f = 1$; then $p^6 = 2^6$ forces $S = \mathrm{PSU}(4, 2)$, which is in the machine base. No other case reaches $p^e = 2^6$. $\square$

6.5. Twisted groups of twisted rank one.

Theorem 6.5. *Let S be one of*

$$\text{PSU}(3, q) \ (q \geq 3), \quad \text{Sz}(q) = {}^2B_2(q) \ (q = 2^f, \ f \text{ odd} \geq 3), \quad {}^2G_2(q) \ (q = 3^f, \ f \text{ odd} \geq 3).$$

Then S is excluded.

Proof. Each group has a single class of parabolic subgroups, namely the Borel subgroups $B = N_S(Q)$, $Q \in \text{Syl}_p(S)$; B is a *maximal* subgroup (rank-one BN -pair) and its class is X -stable for every X . We pair it with a second X -stable maximal class avoiding a Zsigmondy prime.

$\text{PSU}(3, q)$, $q \geq 8$. $|S| = q^3(q^3 + 1)(q^2 - 1)/d$, $d = \gcd(3, q + 1)$; $|B| = q^3(q^2 - 1)/d$. Let V be the stabilizer of a nonisotropic point of the unitary geometry, of order $q(q - 1)(q + 1)^2/d$: maximal for $q \geq 3$, $q \neq 5$ [1, Table 8.5, p. 379] (class C_1), one class, preserved by every semilinear automorphism. Take r a primitive prime divisor of $p^{6f} - 1$ (exists: $p^{6f} = 2^6$ would force $q = 2$, not simple): $r \mid \Phi_6(q) = q^2 - q + 1$, which divides $(q^3 + 1)/(q + 1)$, hence $|S|$; $\text{ord}_r(p) = 6f$ rules out r dividing $q(q^2 - 1)$ or $q(q - 1)(q + 1)^2$. Also $x \mid 2df$, and $r \equiv 1 \pmod{6f}$ gives $r > 2f$, while $r \mid d \mid q + 1$ would force $\text{ord}_r(p) \mid 2f$. So $d_{\text{val}} := v_r(|S|) \geq 1 > 0 = v_r(x)$. For $q \in \{3, 4, 5, 7\}$ (in particular the maximality exception $q = 5$), S is in the machine base.

$\text{Sz}(q)$. $|S| = q^2(q^2 + 1)(q - 1)$; $|B| = q^2(q - 1)$. Let $V := D_{2(q-1)}$, the normalizer of a cyclic subgroup of order $q - 1$: maximal by Suzuki's classification [26]; see also [1, Thm. 7.3.5, p. 367, and Table 8.16, p. 384]. It is one class, stable under $\text{Aut}(S) = S \rtimes C_f$. Let r be a primitive prime divisor of $2^{4f} - 1$ (exists: $4f \geq 12$): $r \mid \Phi_4(q) = q^2 + 1$, so $r \mid |S|$, $r \nmid |B|$, $r \nmid |V|$; $x \mid f$ and $r \equiv 1 \pmod{4f} > f$. So $v_r(|S|) \geq 1 > 0 = v_r(x)$.

${}^2G_2(q)$. $|S| = q^3(q^3 + 1)(q - 1)$; $|B| = q^3(q - 1)$. Let $V := C_S(\iota) \cong \langle \iota \rangle \times \text{PSL}(2, q)$ for an involution ι (one class of involutions, so $[V]$ is Aut -stable), $|V| = q(q^2 - 1)$; V is maximal [16, Thm. C, pp. 33–34]. Let r be a primitive prime divisor of $3^{6f} - 1$ (exists: $6f \geq 18$): $r \mid \Phi_6(q) = q^2 - q + 1$, which divides $(q^3 + 1)/(q + 1)$, so $r \mid |S|$; $\text{ord}_r(3) = 6f$ rules out $r \mid q^3(q - 1)$ and $r \mid q(q^2 - 1)$; $x \mid f < r$. So $v_r(|S|) \geq 1 > 0 = v_r(x)$. (${}^2G_2(3)$ is not simple; ${}^2G_2(3)' \cong \text{PSL}(2, 8)$ is covered by Theorem 6.1.) $\square$

6.6. Untwisted types with graph symmetry.

Theorem 6.6. *Let S be one of*

$$\begin{aligned} &\text{PSL}(n, q) \ (n \geq 3), \quad P\Omega^+(2n, q) = D_n(q) \ (n \geq 4), \quad E_6(q), \\ &\text{Sp}(4, q) \ (q = 2^f, \ f \geq 2), \quad F_4(q) \ (q = 2^f), \quad G_2(q) \ (q = 3^f). \end{aligned}$$

Then S is excluded.

Proof. We keep the standard-parabolic notation Q_J of Lemma 3.6. The *graph part* of X is the image of X in the quotient of $\text{Out}(S)$ by its indiag-field part: a subgroup of $C_2 = \langle \delta \rangle$ for $\text{PSL}(n, q)$ ($n \geq 3$), D_n ($n \geq 5$), E_6 ; a subgroup of S_3 (triality) for D_4 ; while for $\text{Sp}(4, 2^f)$, $F_4(2^f)$, $G_2(3^f)$, $\text{Out}(S)$ is cyclic of order $2f$ generated by the exceptional graph-field automorphism ρ with $\rho^2 = \varphi_p$ [4, Def. 3.1(b), p. 31, and Thm. 3.4, p. 32], and “ X has graph part” means the image of X in $\text{Out}(S)$ is not contained in $\langle \rho^2 \rangle$. An automorphism acts on the building permuting vertex types by a diagram symmetry; automorphisms with trivial graph part preserve every type. By Convention 2.4 we may normalize the graph part up to $\text{Aut}(S)$ -conjugacy; for D_4 this lets us assume a C_2 graph part is generated by the symmetry fixing nodes 1, 2 and swapping 3, 4.

Case A: X with trivial graph part. Every parabolic class is X -stable, and the argument of Theorem 6.3 applies with these data:

- $\text{PSL}(n, q)$: pair $([P_1], [P_2])$, r a primitive prime divisor of $p^{nf} - 1$, so $r \mid \Phi_n(q) \mid q^n - 1$, which divides $|S| \cdot d$ ($d = \gcd(n, q - 1)$); the Levi factors ($\text{GL}_{n-1}(q)$ -type and $\text{GL}_2 \times \text{GL}_{n-2}$ -type) have cyclotomic exponents $\leq (n - 1)f < nf$; $x \mid 2df$ and $r \equiv 1 \pmod{nf}$ rules out $r \mid 2f$ ($r > nf \geq 3f > 2f$) and $r \mid d$ (else $r \mid q - 1$, so $\text{ord}_r(p) \mid f$).

Zsigmondy exceptions $p^{nf} = 2^6$: $(n, f) = (3, 2)$ is $\text{PSL}(3, 4)$ (machine base, all ten X), and $(n, f) = (6, 1)$, $S = \text{PSL}(6, 2)$: take instead $r = 31$ ($\text{ord}_{31}(2) = 5$) and the pair $([P_2], [P_3])$, whose Levi types $\text{GL}_2 \times \text{GL}_4$ and $\text{GL}_3 \times \text{GL}_3$ have exponents $\leq 4 < 5$, so $31 \nmid |P_2||P_3|$ while $v_{31}(|\text{PSL}(6, 2)|) = 1$ and $x \mid 2$.

- $D_n(q)$, $n \geq 4$: pair $([P_1], [P_2])$ (singular point and singular line stabilizers), r primitive for $p^{2(n-1)f} - 1$: $r \mid \Phi_{2n-2}(q)$, which divides $q^{n(n-1)}(q^n - 1) \prod_{i=1}^{n-1} (q^{2i} - 1)$; Levi factors (D_{n-1} - and $A_1 \times D_{n-2}$ -type) have exponents $\leq \max((n-1)f, 2(n-2)f) < 2(n-1)f$; $x \mid 24f$ (diagonal ≤ 4 , field f , graph ≤ 6) and $r \equiv 1 \pmod{2(n-1)f}$ with $2(n-1) \geq 6$ gives $r > 6f$ and $r \nmid 24$ (as r is a prime exceeding 6, $r \notin \{2, 3\}$). Zsigmondy exception $p^{2(n-1)f} = 2^6$: $(n, f) = (4, 1)$, $S = \Omega^+(8, 2)$, handled for all X in Case B3.
- $E_6(q)$: see Case B4; the pair used there is stable for every X .
- $\text{Sp}(4, 2^f)$: pair $([P_1], [P_2])$, r primitive for $2^{4f} - 1$ (exists, $4f \geq 8$): $r \mid \Phi_4(q) = q^2 + 1$; Levi factors $\text{Sp}_2(q)$ and $\text{GL}_2(q)$ have exponents $\leq 2f < 4f$; here $x \mid f$ and $r > 4f$.
- $F_4(2^f)$: pair $([P_1], [P_2])$, r primitive for $2^{12f} - 1$: $r \mid \Phi_{12}(q)$; all proper Levis (types B_3 , C_3 , $A_2 \times A_1$, $A_1 \times A_2$) have exponents $\leq 6f < 12f$; $x \mid f < r$. (This completes the $p = 2$ case excluded from Theorem 6.3.)
- $G_2(3^f)$: pair $([P_1], [P_2])$, r primitive for $3^{6f} - 1$: $r \mid \Phi_6(q)$; Levis of type A_1 have exponents $\leq 2f < 6f$; $x \mid f < r$. (The $p = 3$ case of G_2 .)

Case B: X with nontrivial graph part. Novelty classes below are standard parabolics Q_J handled by Lemma 3.6: each $[Q_J]$ is X -stable because J is invariant under the relevant diagram symmetry (an automorphism with graph part γ maps $[Q_J]$ to $[Q_{\gamma(J)}]$), and its proper parabolic overgroups fall into nontrivial γ -orbits, giving hypothesis (b) of Lemma 3.6; so B_{Q_J} is maximal in G by Lemma 3.4, and pairs of distinct classes are non-conjugate by Lemma 3.5. The primes r are those of Case A unless stated; r -avoidance for Q_J follows from its own Levi order (a product of cyclotomic factors of the Levi of type J and powers of $q - 1$), as listed.

B1. $\text{PSL}(3, q)$, $q \geq 5$ ($q \leq 4$: machine base, including all ten X for $\text{PSL}(3, 4)$). Pair: $[B]$ (novelty; the proper overgroups P_1, P_2 are swapped by δ , which exchanges points and lines of $PG(2, q)$) and $[N]$, $N := N_S(T) \cong ((q-1)^2/d) \cdot S_3$ the normalizer of a maximal split torus ($d = \gcd(3, q-1)$), maximal for $q \geq 5$ [20, 12], [1, Table 8.3, p. 378] (class C_2), one class, preserved by every automorphism. Take r primitive for $p^{3f} - 1$: $r \mid \Phi_3(q) = q^2 + q + 1$; $r \nmid |B| = q^3(q-1)^2/d$ (primitivity) and $r \nmid |N| = 6(q-1)^2/d$ (as $r \equiv 1 \pmod{3}$ forces $r \geq 7 > 6$, $r \nmid 6$); $v_r(|S|) \geq 1$. Also $x \mid 2df$ and $r > 3f$ rules out $r \mid 2f$ and $r \mid d = 3$. Zsigmondy exception $p^{3f} = 2^6$ is $\text{PSL}(3, 4)$: machine base.

B2. $\text{PSL}(n, q)$, $n \geq 4$. For $n = 4$: pair $([P_2], [Q_{\{2\}}])$. Node 2 is δ -fixed, so $[P_2]$ is an X -stable maximal class (Levi $A_1 \times A_1$, exponents $\leq 2f$); $Q_{\{2\}}$ is the stabilizer of an incident point-hyperplane flag, with proper overgroups P_1, P_3 swapped by δ (Lemma 3.6); its Levi (A_1 plus torus) has exponents $\leq 2f$. Take r primitive for $p^{4f} - 1$: it avoids both $(2f < 4f)$, $v_r(|S|) \geq 1$, and $x \mid 2df$ ($d = \gcd(4, q-1)$) with $r \equiv 1 \pmod{4f}$ excluding $r \mid 2f$ and $r \mid d$. No Zsigmondy exception ($2^{4f} = 2^6$ impossible). For $n \geq 5$: pair of novelties $([Q_{J_1}], [Q_{J_2}])$, $J_1 := \Delta \setminus \{1, n-1\}$, $J_2 := \Delta \setminus \{2, n-2\}$: the stabilizers of an incident point-hyperplane flag and an incident line- $(n-2)$ -space flag. Both node sets $\{1, n-1\}$, $\{2, n-2\}$ are δ -orbits, so both $[Q_{J_i}]$ are X -stable; the proper overgroups of Q_{J_1} are P_1, P_{n-1} (swapped by δ), of Q_{J_2} are P_2, P_{n-2} (swapped, distinct classes since $n \geq 5$). Levi types: blocks $(1, n-2, 1)$, i.e. GL_{n-2} plus torus, exponents $\leq (n-2)f$; and blocks $(2, n-4, 2)$, exponents $\leq \max(2, n-4)f$. The prime r of Case A avoids both, and $v_r(x) = 0$ as there. Zsigmondy exception $(n, f) = (6, 1)$, $S = \text{PSL}(6, 2)$: take $r = 31$ again: the two Levi orders involve only $\text{GL}_4(2)$, $\text{GL}_2(2)$ factors and 2-powers, so 31 divides neither, while $v_{31}(|S|) = 1 > 0 = v_{31}(2)$.

B3. $D_n(q)$. For $n \geq 5$ the graph part is at most the C_2 swapping the two spinor nodes $n-1, n$ and fixing $1, \dots, n-2$: the Case A pair $([P_1], [P_2])$ is X -stable for *every* X and Case A's argument goes through unchanged (its x -bound already allowed the graph factor). For $n = 4$: if the graph part is trivial or the normalized C_2 (fixing nodes $1, 2$), the pair $([P_1], [P_2])$ is X -stable and Case A applies. If the graph part contains a triality, take the pair $([P_2], [Q_{\{2\}}])$: node 2 is fixed by all of S_3 , so $[P_2]$ is an X -stable maximal class (Levi $A_1 \times A_1 \times A_1$, exponents $\leq 2f$), and $Q_{\{2\}}$ (Levi A_1 plus torus, exponents $\leq 2f$) has proper overgroups $Q_{\{1,2\}}, Q_{\{2,3\}}, Q_{\{2,4\}}$ and P_4, P_3, P_1 , which fall into the two regular (triality)-orbits induced by the permutation of the end nodes $\{1, 3, 4\}$; none is X -stable (Lemma 3.6). The prime r primitive for $p^{6f} - 1$ avoids both classes, $v_r(|S|) \geq 1$, and $r \nmid x$ as in Case A. *Zsigmondy exception* ($f = 1, p = 2$): $S = \Omega^+(8, 2)$. Here $x \mid 24$; take $r = 5 = \Phi_4(2)$, so $v_5(|S|) = v_5(2^{12}(2^6 - 1)(2^4 - 1)^2(2^2 - 1)) = 2$ and $v_5(x) \leq v_5(24) = 0$. Triality case, pair $([P_2], [Q_{\{2\}}])$: both Levi orders are $\{2, 3\}$ -numbers, so $d = 2 - 0 - 0 = 2 > 0$. Trivial/ C_2 case, pair $([P_1], [P_2])$: $v_5(|P_1|) = v_5(|\mathrm{SL}_4(2)|) = 1$ (Levi $D_3 \cong A_3$), $v_5(|P_2|) = 0$, so $d = 2 - 1 - 0 = 1 > 0$.

B4. $E_6(q)$, all X . The diagram involution fixes nodes 2 and 4 (Bourbaki numbering: δ swaps $1 \leftrightarrow 6, 3 \leftrightarrow 5$). Pair $([P_2], [P_4])$: two X -stable classes of *maximal* parabolics for every X . r primitive for $p^{12f} - 1$: $r \mid \Phi_{12}(q)$, which divides $|S| \cdot d$; the Levi of P_2 has type A_5 (exponents $\leq 6f$), that of P_4 type $A_2 \times A_1 \times A_2$ ($\leq 3f$), both $< 12f$; $x \mid 2df \leq 6f < r$ ($d = \gcd(3, q - 1)$). No Zsigmondy exception. This simultaneously covers Case A for E_6 .

B5. $\mathrm{Sp}(4, q)$, $q = 2^f$, $f \geq 2$, X with graph part. ($f = 1$: $\mathrm{Sp}(4, 2)$ is not simple; its derived group A_6 is in the machine base.) Pair: $[B]$ (novelty: the proper overgroups P_1, P_2 (the point and totally-isotropic-line stabilizers) are swapped by ρ ; Lemma 3.6) and $[V]$, where $V := \mathrm{Fix}(\tilde{\rho}^f) \cap S$ for $\tilde{\rho}$ the Steinberg endomorphism of the ambient algebraic group with $\tilde{\rho}^2 = \varphi_2$ and $\tilde{\rho}^{2f} = F_q$. Explicitly $V = \mathrm{Sz}(q)$ for f odd and $V = \mathrm{Sp}(4, 2^{f/2})$ (subfield subgroup, index-two field extension) for f even; V is maximal in both cases [1, Table 8.14, p. 384], [26]. *Stability of $[V]$* : $\mathrm{Aut}(S) = \mathrm{Inn}(S)\langle \rho \rangle$ with $\rho = \tilde{\rho}|_S$; $\tilde{\rho}$ commutes with its own power $\tilde{\rho}^f$, so $\rho(V) = V$, and inner automorphisms preserve $[V]$ trivially; hence $[V]$ is Aut-stable. *Primes*: f odd, $2f \neq 6$: r primitive for $2^{2f} - 1$, so $r \mid q + 1$: $v_r(|S|) = v_r((q^2 - 1)(q^4 - 1)) = 2v_r(q + 1) \geq 2$; $v_r(|B|) = v_r(q^4(q - 1)^2) = 0$; $v_r(|\mathrm{Sz}(q)|) = 0$ since $\gcd(q + 1, q(q^2 + 1)(q - 1)) = 1$ for q even; $x \mid 2f < r$. If $2f = 6$ ($q = 8$): $r = 3$ gives $v_3(|\mathrm{Sp}(4, 8)|) = v_3(63 \cdot 4095) = 4 > 1 \geq v_3(x)$, with $v_3(|B|) = v_3(|\mathrm{Sz}(8)|) = 0$. f even: r primitive for $2^{4f} - 1$ (exists, $4f \geq 8$), $r \mid q^2 + 1$: $v_r(|S|) \geq 1$; $|\mathrm{Sp}(4, q_0)| = q_0^4(q_0^2 - 1)(q_0^4 - 1)$ with $q_0^2 = q$ has cyclotomic exponents $\leq 2f < 4f$, so $v_r(|V|) = 0$; $v_r(|B|) = 0$; $x \mid 2f < r$. ($\mathrm{Sp}(4, 4)$ doubles as the machine template: the sweep-K3 certificate pair (Borel, $\mathrm{Sp}(4, 2)$) at $p = 5 = \Phi_4(2)$ and $p = 17 = \Phi_8(2)$ matches this construction exactly.)

B6. $F_4(q)$, $q = 2^f$, X with graph part. ρ reverses the diagram $1-2 \Rightarrow 3-4$ (swaps $1 \leftrightarrow 4, 2 \leftrightarrow 3$). Pair of novelties $([Q_{J_1}], [Q_{J_2}])$, $J_1 := \{2, 3\}$, $J_2 := \{1, 4\}$, both ρ -invariant node sets. Proper overgroups of Q_{J_1} : $Q_{\{1,2,3\}} = P_4$ and $Q_{\{2,3,4\}} = P_1$, swapped by ρ ; of Q_{J_2} : $Q_{\{1,2,4\}}$ and $Q_{\{1,3,4\}}$, swapped by ρ (distinct types, hence distinct classes). Lemma 3.6 applies to both. Levi types: $B_2 (\cong \mathrm{Sp}_4(q))$, exponents $\leq 4f$ and $A_1 \times A_1 (\leq 2f)$. r primitive for $2^{12f} - 1$ avoids both; $v_r(|S|) \geq 1$; $x \mid 2f < 12f < r$. No exceptions.

B7. $G_2(q)$, $q = 3^f$, X with graph part. ρ swaps the two nodes (long and short roots), $\rho^2 = \varphi_3$. Pair: $[B]$ (novelty: overgroups P_1, P_2 swapped by ρ ; Lemma 3.6) and $[V]$, $V := \mathrm{Fix}(\tilde{\rho}^f) \cap S$ exactly as in B5: $V = {}^2G_2(q)$ for f odd, maximal in $G_2(q)$ [16, Thm. A, p. 33] (for $f = 1$, $V = {}^2G_2(3) \cong \mathrm{P}\Gamma\mathrm{L}_2(8)$, maximal in $G_2(3)$ by the ATLAS [6] and [16, Thm. A, p. 33]), and $V = G_2(3^{f/2})$ for f even (subfield, index-two extension, maximal [16, Thm. A, p. 33]). $[V]$ is Aut-stable by the commuting-endomorphism argument of B5 ($\mathrm{Out}(G_2(3^f)) = \langle \rho \rangle \cong C_{2f}$). *Primes*: f odd: r primitive for $3^{3f} - 1$ (exists for all $f \geq 1$: $p = 3 \neq 2$ excludes the 2^6 exception; $f = 1$ gives $r = 13 \mid 3^3 - 1$). Then $r \mid \Phi_3(q) = q^2 + q + 1$, which divides $q^6 - 1$, hence

$|S| = q^6(q^6 - 1)(q^2 - 1)$; $r \nmid |B| = q^6(q - 1)^2$ (primitivity); $r \nmid |{}^2G_2(q)| = q^3(q^3 + 1)(q - 1)$ since $r \mid q^3 - 1$ and $\gcd(q^3 - 1, q^3 + 1) = 2 < r$, while $r \nmid q - 1$ by primitivity ($r \mid 3^f - 1$ would force $\text{ord}_r(3) \leq f < 3f$); $x \mid 2f$ and $r \equiv 1 \pmod{3f}$ gives $r \geq 3f + 1 > 2f$, so $v_r(x) = 0 < 1 \leq v_r(|S|)$. f even: r primitive for $3^{6f} - 1$, so $r \mid \Phi_6(q) = q^2 - q + 1 \mid q^3 + 1$; $r \nmid |B|$; $r \nmid |G_2(3^{f/2})| = q^3(q^3 - 1)(q - 1)$ (as $r \mid q^3 + 1$); $x \mid 2f < 6f < r$. ($G_2(3)$ additionally carries a full machine certificate for both X .)

In every case Theorem 4.1 excludes S^k for every $k \geq 2$. $\square$

7. PROOF OF THE MAIN THEOREM

Proposition 7.1 (Coverage). *Every non-abelian finite simple group is excluded in the sense of Convention 4.4.*

Proof. By the classification of finite simple groups [10, Ch. 1, Table I, pp. 8–10], S is alternating, of Lie type, or one of the 26 sporadic groups; we run through the list and name the theorem responsible.

- A_n ($n \geq 5$): Theorem 6.2.
- $\text{PSL}(2, q)$ ($q \geq 4$): Theorem 6.1.
- $\text{PSL}(n, q)$ ($n \geq 3$), $D_n(q)$ ($n \geq 4$), $E_6(q)$: Theorem 6.6.
- $\text{PSp}(2n, q)$: $n \geq 3$, or $n = 2$ with q odd: Theorem 6.3; $\text{Sp}(4, 2^f)$, $f \geq 2$: Theorem 6.6 ($\text{Sp}(4, 2)$ is not simple).
- $\Omega(2n+1, q)$, q odd, $n \geq 3$: Theorem 6.3 (for q even, $B_n(q) \cong C_n(q)$, already listed; $\Omega(5, q) \cong \text{PSp}(4, q)$).
- $E_7(q)$, $E_8(q)$; $F_4(q)$, $p \neq 2$; $G_2(q)$, $p \neq 3$: Theorem 6.3. $F_4(2^f)$ and $G_2(3^f)$: Theorem 6.6. ($G_2(2)$ and ${}^2G_2(3)$ are not simple; their derived groups $\text{PSU}(3, 3)$ and $\text{PSL}(2, 8)$ are covered.)
- $\text{PSU}(n, q)$: $n = 3$: Theorem 6.5; $n \geq 4$: Theorem 6.4.
- ${}^2D_n(q)$ ($n \geq 4$), ${}^3D_4(q)$, ${}^2E_6(q)$, ${}^2F_4(2^f)'$: the first three and ${}^2F_4(2^f)$ (f odd ≥ 3) by Theorem 6.4; the Tits group ${}^2F_4(2)'$ by Proposition 5.2. (2D_2 , 2D_3 are $\text{PSL}(2, q^2)$, $\text{PSU}(4, q)$, already listed.)
- $\text{Sz}(2^f)$ (f odd ≥ 3), ${}^2G_2(3^f)$ (f odd ≥ 3): Theorem 6.5.
- Sporadic groups: Proposition 5.2.

Every finite simple group appears (repetitions from exceptional isomorphisms are harmless), so the proposition follows. $\square$

Proof of Theorem 1.1. Suppose not, and let G be a non-soluble group with property P of least order. By Proposition 2.3, G has a unique minimal normal subgroup $N = S^k$, S non-abelian simple, and after conjugation $N \leq G \leq X \wr S_k$ as in Convention 2.4. If $k = 1$, then G is almost simple, contrary to the theorem of Tikhonenko–Tyutyanov [28]. Hence $k \geq 2$. By Proposition 7.1 there are classes $[V] \neq [W]$, maximal in $\mathcal{P}_X(S)$ and normally saturating, and a prime p with $v_p(|S|) - v_p(|V|) - v_p(|W|) > v_p(x)$. By Theorem 4.1, G does not have property P, a contradiction. $\square$

Corollary 7.2 (Supplementary reproof). *The present criterion also recovers that no almost simple group has property P.*

Proof. Theorem 4.3 with the same witnesses used in Proposition 7.1. This corollary is not used in the proof of the Main Theorem. $\square$

Remark 7.3. The proof yields slightly more than solubility of the minimal counterexample: for every finite group G with a unique minimal normal subgroup S^k (S non-abelian simple, $k \geq 2$),

some pair of non-conjugate maximal subgroups of G fails to factorize G . The reduction of §2 is needed only to pass from an arbitrary non-soluble group to this configuration.

Data Availability. All code, GAP scripts, and log certificates required to reproduce the machine-verified claims of §5, together with the arithmetic receipts for the family proofs of §6, are archived in the versioned release [24]. The release includes SHA-256 checksums of every certificate log, a CITATION.cff file, a single command running the quick verification suite (coverage cross-checks, independent finite-witness and source-map checks, symbolic family checks, manuscript/audit lint, and hash verification; no GAP recomputation required), and separate documented commands for the full GAP, Lean, clean-diff, and mutation-test reproduction.

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